Raghuvendra Kumar

AI Engineer — Generative AI, ML & LLM Applications

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Summary

AI/ML engineer building production-grade Generative AI systems — RAG, multi-agent architectures, and speech-to-text pipelines — with LangChain, LangGraph, and FAISS/ChromaDB. Shipped 3 end-to-end GenAI applications across healthcare, research automation, and multimodal retrieval, each deployed and publicly demoable. Solved 373+ problems on LeetCode.

Education

Vellore Institute of Technology — **Expected 2027**
B.Tech. in Artificial Intelligence and Machine Learning — *Bhopal, MP*

DAV Public School — **2020 – 2023**
Higher Secondary Education (CBSE) — *Gaya, Bihar*

Technical Skills

Languages: Python, Java, SQL, JavaScript
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy, OpenCV, Streamlit, Sentence-Transformers
Generative AI, NLP & Agentic Systems: LangChain, LangGraph, RAG, FAISS, ChromaDB, Whisper, Gemini API, Mistral AI, MCP, FastMCP
Frontend, Backend, Cloud & Tools: HTML/CSS, FastAPI, AWS, PostgreSQL, SQLite, Docker, Git, GitHub, GitHub Actions

Projects

OncoGuide AI — Evidence-Grounded Medical Report Assistant — GitHub | Live Demo | Demo Video
Python, LangChain, Gemini API, FAISS, Sentence-Transformers, Streamlit

- Engineered a RAG-based chatbot enabling cancer patients to query 6 report types (pathology, MRI, CT, PET, blood, follow-up) in plain English, with automatic report classification and OCR fallback for scanned PDFs.
- Designed a retrieval pipeline (PyMuPDF extraction, semantic chunking, Sentence-Transformer embeddings, FAISS similarity search) that grounds every response in retrieved evidence, reducing hallucination risk in clinical Q&A.
- Delivered cross-report reasoning, conversational memory, automated patient timelines, and structured summaries; migrated the backend from local Ollama/Llama 3 to the Gemini API, cutting deployment overhead for production use.

Multi-Agent Research System — LangGraph Research Agents — GitHub | Live Demo | Demo Video
LangChain, LangGraph, OpenRouter (Llama 3.3 70B), Tavily, Streamlit

- Architected a 4-agent system (Search, Reader, Writer, Critic) with LangGraph and Llama 3.3 70B, orchestrating end-to-end web research, content synthesis, and automated quality review.
- Integrated the Tavily API for real-time multi-source retrieval, enabling agents to ground synthesized reports in live web data rather than static knowledge.
- Generated structured, 1,000+ word research reports via iterative agent-based review and prompt engineering; deployed and load-tested on Streamlit Cloud.

AI Video Assistant — Whisper + RAG + Mistral AI — GitHub | Live Demo | Demo Video
OpenAI Whisper, Sarvam AI, Mistral AI, LangChain, RAG, ChromaDB, Streamlit

- Built a 6-stage AI pipeline (extraction, preprocessing, chunking, transcription, embedding, RAG retrieval) converting audio/video into transcripts and conversational answers.
- Implemented dual STT engines (OpenAI Whisper, Sarvam AI) with ChromaDB-backed retrieval, enabling context-preserving chat across English, Hindi, and Hinglish.
- Shipped a Streamlit app supporting uploads up to 200MB, turning meetings, podcasts, and lectures into searchable transcript-based Q&A.

Other ML Projects: Plant Disease Classification (CNN, TensorFlow/Keras, 86% accuracy on 54K images) — Customer Churn Prediction (XGBoost, SMOTE, 79.5% accuracy)

Certifications

Applied Machine Learning in Python — University of Michigan, Coursera — Nov 2025
AWS Cloud Practitioner — GeeksforGeeks — Oct 2025
TensorFlow Course — GeeksforGeeks — Sep 2025
Cloud Computing — NPTEL — May 2025